OSPF Day-28

Nov-29

OSPF Part-III

Loopback Interfaces

- A loopback interface is the virtual interface in the router.
- It's always up/up (unless we manually shut it down)
- It's status isn't dependent on physical interface. Physical int. can fail due to hardware problems however loopback int. doesn't fail unless router itself fail
- So it provides consistent IP add. that can be used to reach/identify the router.

Sometimes we need to send traffic directly to router. Suppose R1 has no loopback interface & R4 receives a packet destined for R1 at 10.0.13.1, the IP address of its G1/0 int. It might forward it via R3 R3. If R1 G2/0 interface goes down due to some reason, R4 won't be able to send packets to R1. However if R1 has loopback add 1.1.1.1 & is used to identify R1. Even if physical int (of R1) fails, if R4 receives packet destine for R1 loopback interface it will still be able to send packet to R1. So, its good idea to configure loopback interface on router.

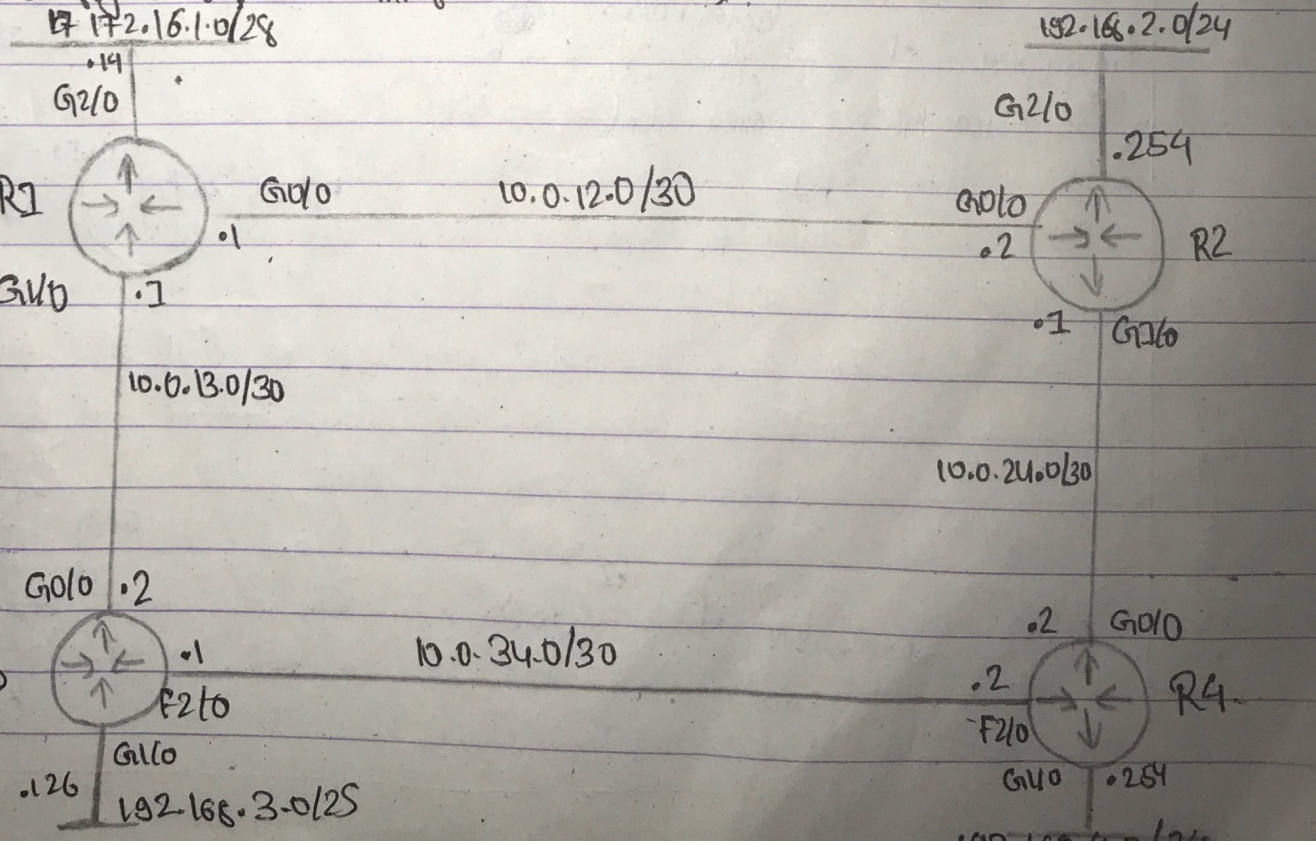


Fig Network-I

OSPF Network Types

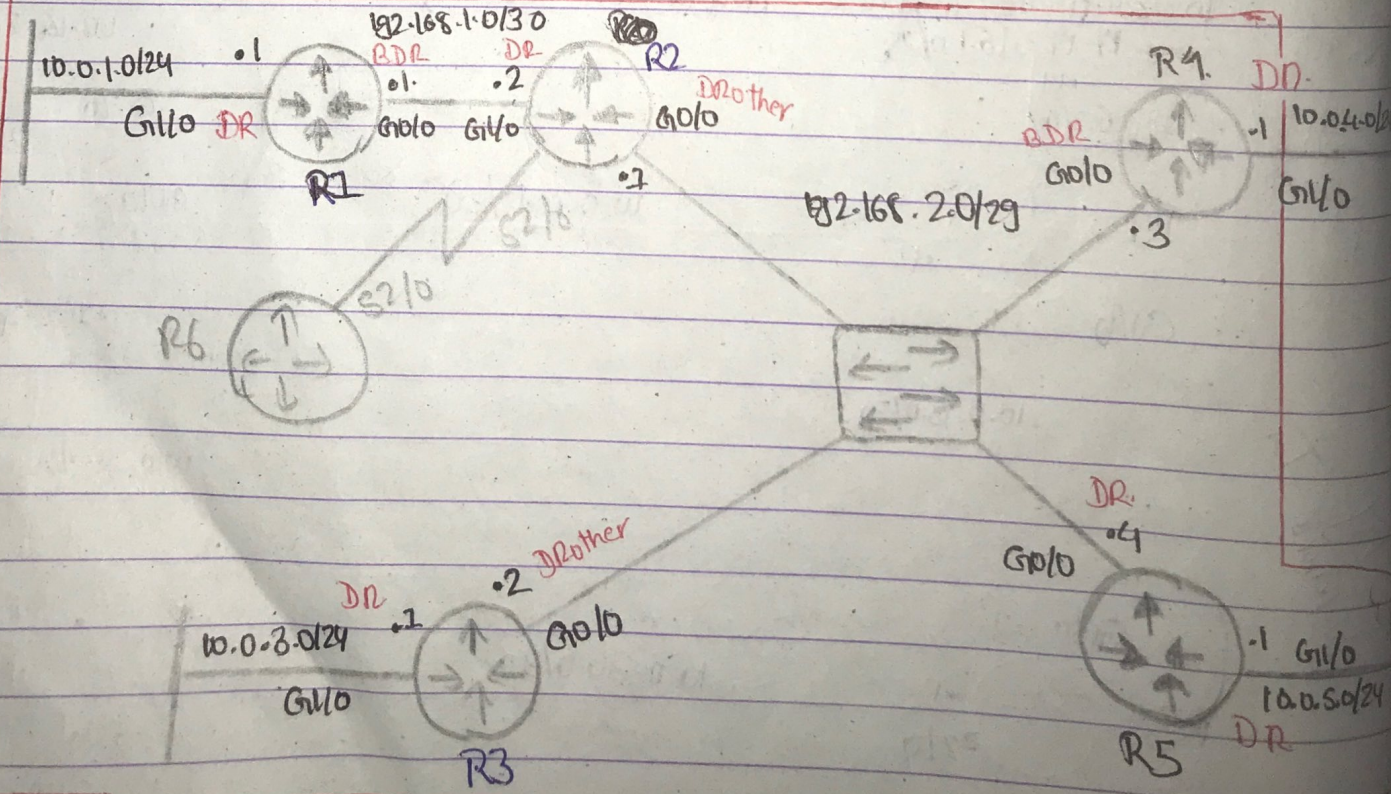
The OSPF (network type) refers to the type of connection between OSPF neighbors (ethernet, etc)

- There are three main OSPF network types.

1) Broadcast. (Focus for CCNA as its in syllabus)
   enabled by default on Ethernet & FDDI (Fiber Distributed Data Interfaces).

2) Point-to-Point. (CCNA syllabus has it)
   enabled by default on PPP (Point-to-Point Protocol) and HDLC (High Level Data Link Control) interfaces

3) Non-broadcast.
   enabled by default on Frame Relay and X.25 interfaces.



HW - II
Network - II

1) Broadcast Network type.

- Enabled on Ethernet and FDDI interfaces by default.
- Routers dynamically discover neighbors by sending/listening for OSPF Hello messages using multicast address 224.0.0.5
- A DR a(designated router) and BDR (Backup Designated Router) must be elected on each subnet(only DR if there are no OSPF neighbors, ie R1's G1/0 interface).
- Routers which aren't the DR or BDR become DROther.

DR/BDR priority order.

- The DR/BDR election order of priority:

1: Highest OSPF interface priority. (all have same by default).
2) Highest OSPF Router ID.

- First place becomes DR for subnet, second place' becomes the BDR.
- # The default OSPF inteface priority is 1 on all interfaces. On our network-II, for 192.168.02.0/29 subnet we'd set up loopback int on all routers. So, Router ID of R5 was gr highest and it became DR 4

Changing OSPF int priority.

Now, what if we want R2 to be DR instead of R5. Lets see changing OSPF int priority.

The command is given below:

R

R2(config)# int g0/0

R2(config-if)# ip ospf priority 255.
<0-255> priority

NOB: If we set the OSPF interface priority to 0, the router CANNOT BE DR/BDR o for the subnet.

Now, if we check ospf status with sh ip ospf int g0/0 on R2 we'll find R2 is still DROther instead of being DR.

The DR/BDR election is 'non-preemptive'. Once the DR/BDR are selected they will keep their role until OSPF is reset, the int fails/is shut down etc.

R5

Now, if we reset ospf with #, clear ip ospf process & check once more, we'll observe following things.

- R4 become the DR, not R2. R2 became BDR. (though R2 has highest priority)
  4) When the DR goes down, the BDR becomes the new DR. Then an election is held for next BDR.

- R3 is DR other & stable in 2way state. (also R5)
  DR other will only move to full state with DR & BDR.
  The neighbor state with other DROthers will be 2-ways

- On the broadcast network type, routers will only form a Full OSPF adjacency with the DR and BDR of the segment.

- Therefore, routers only exchange LSAs with DR and BDR, DROthers willn't exchange LSAs with each other.

- All routers will still have the same LSDB, but this reduces the amount of LSAs flooding the network.

~~Mssg~~
NOB: Mssg to DR/BDR are multicast using address 224.0.0.6
The DR and BDR will form a FULL Adjacency with ALL routers on subnet.
DROthers will form FULL Adjacency only with the DR/BDR.

gn#sh ip ospf int brief command. there is col Nbrs F/C.
F indicates no-of full adjacencies & C indicates count of neighbors

2) Point-to-Point Network Type.

- Enabled on Serial Interfaces using the PPP or HDLC encapsulations by default.
- Routers dynamically discover neighbors by sending/listening for OSPF Hello messages using multicast address 224.0.0.5.
- A DR and BDR are not elected.
- These encapsulations are used for 'point-to-point' connections.
- Therefore there is no point in electing a DR & BDR.
- The routers will form full adjacency with each other.

Serial connections are old technology which isn't very common anymore. They still exist but Internet is much more dominant nowadays.
- Default Timers: Hello 10, Dead 40

Scanned with CamScanner

Serial Interfaces

```
R1(config)# int s2/0
R1(config-if)# clock rate 64000 (bits in bits per second)
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

- One side of a serial connection functions as DCE (Data Communications Equipment)
- The other side functions as DTE (Data Terminal Equipment)
- The DCE side needs to specify clock rate (speed) of the interface. In our network R1 has DCE side of cable & therefore needs to tell R6, what speed connection will operate at.

NOB:

Ethernet interfaces use the speed command to configure the interfaces operating speed. Serial interfaces use the clock rate command.

- The default encapsulation on a serial interface is HDLC.
  ** actually cHDLC (Cisco HDLC).
  (the serial connection has doesn't use mac address).

- To change encapsulation to PPP, lets use the following command.

```
R1(config-if)# int s2/0
R1(config-if)# do sh int s2/0 , encapsulation ppp
                Encapsulation PPP
```

- If we change the encapsulation, it must match on both ends or the interface will go down.

DCE & DTE -

To see if router's int is in serial connection is DTE or DCE use the command below.

R1 # show controllers s2/0

* In point-to-point there is no DR or BDR.

Configure the OSPF Network Type.

We can manually configure network type and interface uses, with following commands:

R1(config-if) # ip ospf network ?

broadcast
non-broadcast
point-to-multipoint
point-to-point

ip ospf network type.

• For eg, if two routers are directly connected with an Ethernet link, there is no need for a DR/BDR. We can configure point-to-point network type in this case.

NOB:

Not all network types work on all link types (for example: a serial link can't use broadcast network type as it doesn't support layer 2 broadcast frames, which is necessary for broadcast network type).

Non-broadcast network type default timers = Hello 30, Dead 120

OSPF Neighbor Requirements

1) Area ~~must~~ number must match.
2) Interfaces must be in same subnet.
3) OSPF process mustn't be shutdown.
   R1(config)~~router~~) # router ospf 1
   R1(config-router) # shutdown
   R1 (config-router) # no shutdown.

4) OSPF router IDs must be unique.
   We can clear router id with below command
   R1 (config-router) no router-id

5) Hello & Dead Timers must match.
   Hello & dead timers are directly configured on interface.

   R2(config-if) # ip ospf hello-interval 5 <seconds>
   R2(config-if) # ip ospf hello dead-interval 10 <seconds>

   R2(config-if) # no ip ospf hello/dead-interval
   (This command returns it to default values)

6) Authentication settings must match.
   We can configure OSPF password & routers will only become neighbors with routers that have matching OSPF password.

- R2(config-if) # ip ospf authentication-key ~~pass~~ that123
  (this doesn't enable authentication, password is only configured)
- R2(config-if) # ip ospf authentication
  (this enables authentication)

Scanned with CamScanner

Similarly no ip ospf authentication-key jeremy & no ip ospf authentication

7) IP MTU settings must match
Even if settings don't match, can become OSPF neighbor, but it doesn't operate properly
R2(config-if)# ip mtu ?
<68-1500> in byte)
R2(config-if)# ip mtu 1400.
(default is 1500)

8) OSPF network Type must match

OSPF LSA Types.
The OSPF LSDB is made up of LSAs.

- There are 11 types of LSA, but there are only 3 you should be aware of for the CCNA.
  Type 1 (Router LSA)
  Type 2 (Network LSA)
  Type 5 (AS External LSA)

#) Type 1 (Router LSA)
- Every OSPF router generates this type of LSA.
- It identifies the router using its router ID
- It also lists network attached to router's OSPF-activated interfaces.

Scanned with CamScanner

Type 2 (Network LSA)

- Generated by DR of each 'multi-access' network (ie broadcast network type)
- List the routers which are attached to multi-access network.
- ~~Type 5 (AS-External LSA)~~
  - Grena

Type 5 (AS-External LSA)

Generated by ASBRs to describe routes to destinations outside of AS (OSPF domain)

Scanned with CamScanner